

## Binary Distillation

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## Introduction

- The objective of this experiment is to determine the boil-up rates, the column efficiency, the column pressure drop and the product composition at different boiler power and reflux ratios. The McCabe-Thiele method is used to determine their parameters.

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## Theory

- In a typical batch distillation column, separation is done by gravity where the liquid falls and the vapor rises. The feed enters near the middle of the column. Heat is supplied to the reboiler, so it acts as a heat source to boil the mixture in the column. Stages where the liquid and vapor contact are present within the column, allowing the liquid to fall down the column and the vapor to rise.
- The McCabe-Thiele analysis for binary distillation is used to design distillation columns but it requires many assumptions. The McCabe-Thiele method assumes vapor-liquid equilibrium at each stage, but this is a poor assumption because equilibrium is never obtained completely in practice. Efficiency is then introduced with the Murphree plate efficiency equation.

$$E_M = \frac{y_n - y_{n+1}}{y_n^* - y_{n+1}}$$

$y_n$  = composition of vapor leaving stage  $n$   
 $y_{n+1}$  = composition of liquid entering stage  $n+1$   
 $y_n^*$  = composition of vapor in equilibrium with  $y_{n+1}$

$$N = \frac{\ln \left( \frac{y_1 - y_2}{y_2 - y_1^*} \right)}{\ln \left( \frac{y_1 - y_2}{y_2 - y_1^*} \right)}$$

$N$  = number of theoretical plates  
 $N_{actual}$  = number of actual plates  
 $E_M = \frac{N}{N_{actual}}$

Average relative volatility (assumed constant) =  $\alpha_{avg}$   
 $y_n = \alpha_{avg} x_n$  if the vapor and liquid are in thermodynamic equilibrium

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## Apparatus



Main apparatus has Reboiler, Heat Exchanger condenser, bottom product receiver, column, top receiver, feed tank, decanter, feed pump, monometer, reflux valve, cooling water valve, control panel, computer.



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## Slide 1

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**DL1** 97/100

Donglin Li, 11/22/2019

## Slide 3

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**DL2** Type the equations.

Donglin Li, 11/22/2019

**DL3** -1

Donglin Li, 11/22/2019

**DL4** Too small the font size. Could divide this slide into 2 or more slides.

Donglin Li, 11/22/2019

**DL5** -1

Donglin Li, 11/22/2019

## Slide 4

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**DL6** State the name of the valves clearly (V2, V3, V5, V6, V7 and V10).

Donglin Li, 11/22/2019

**DL7** -1

Donglin Li, 11/22/2019

## Material & Supplies

- Materials used were Ethanol and DI-Water for mixture
- Graduated cylinder to measure flow rate
- Refractometer to measure composition of mixture
  - Cleaning refractometer with tissue and di-water
- Test tubes were used to collect mixtures and cooled in Bath covered with parafilm
- Water Bath for test tubes to maintain at room temperature
- Use Pipets for transfer mixture to Refractometer
- Clean area frequently

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## Procedure

- Add ethanol and water mixture to reboiler (10L of 10wt% ethanol/water)
  - Heat until temperature profile becomes constant
- Obtain calibration curve using different compositions of water/ethanol
- Experiment A: Effect of boil-up rate and overall column efficiency
  - Collect 3 samples of distillate and bottom composition for 4 different boil-up rates provided by the heat controller at 0.50kW, 0.75kW, 1.00kW and 1.25kW. Collect pressure drop as well. Collect the composition of each sample using the pocket refractometer.
- Experiment B: Distillation at Constant Reflux Ratio
  - Set heat controller to 0.75kW, wait for 15 min to allow the system to reach equilibrium
  - The reflux ratio was changed and the same procedure as experiment A was done for collecting distillate and bottoms, the only difference is that the heat controller stayed at 0.75kW

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## Experimental Challenges

- Make sure to not reuse pipets and wash the graduated cylinder after each collection to reduce contamination
- Label test tubes and use parafilm to reduce evaporation after collection
- Make sure to save the temperature collection data from the computer after the lab
- The liquids will be hot, use insulated gloves if too hot

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## Safety

- PPE
  - Safety Glasses
  - Gloves
  - Heat protective gloves
- Do careful with handling high concentration Ethanol composition
  - Ethanol is a OSHA Flammable None Halogenated Liquid

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